

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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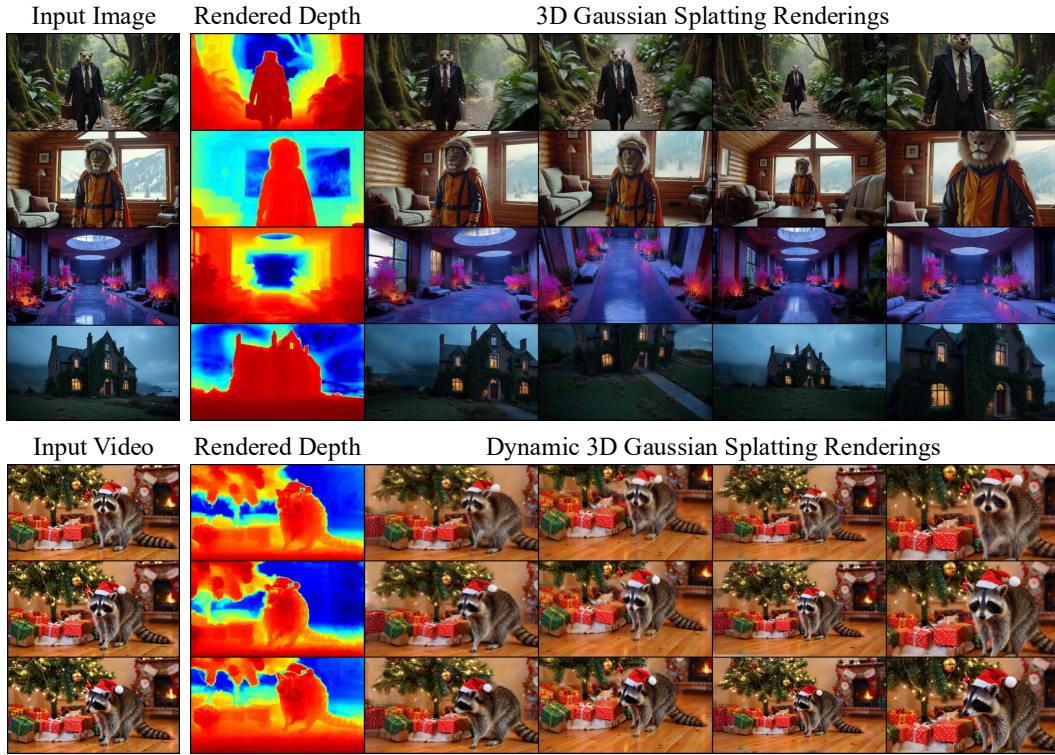


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* predicts a 3D Gaussian Splatting (3DGS) representation in one feed-forward pass by self-distilling a video diffusion model, without using real-world multi-view training data. From a video input (bottom), *Lyra* predicts a dynamic 3DGS whose renderings can be compared across both time (rows) and viewpoint (columns), illustrating the same controllability evaluated quantitatively in the experiments.

ABSTRACT

Generating virtual environments at scale is increasingly important for applications ranging from gaming to physical AI domains such as robotics, autonomous driving, and industrial AI. Learning-based 3D reconstruction methods can produce explicit geometry, but their empirical strength is tied to captured real-world multi-view data, which is costly, unevenly available, and especially difficult to obtain for dynamic scenes. Video diffusion models provide a complementary source of evidence: they are trained on broad video corpora and show strong 2D imagination, yet their frame-based outputs do not directly support simulation settings where agents must navigate, render from new viewpoints, and interact with a consistent environment. We propose a self-distillation framework that converts the implicit 3D knowledge of video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, removing the need for multi-view training captures. Concretely, we augment the standard RGB decoder with a 3DGS decoder and supervise the 3DGS branch using the RGB branch outputs produced by the same video latent. This design lets the 3DGS decoder train purely from synthetic supervision generated by

video diffusion models while preserving the comparison target, camera condition, and rendered-view metric structure of reconstruction evaluation. At inference time, the model synthesizes 3D scenes from either a text prompt or a single image and supports real-time rendering. The same framework further extends to dynamic 3D scene generation from a monocular input video. Across static and dynamic 3D scene generation benchmarks, the experiments show state-of-the-art quantitative performance, with consistent gains over prior methods across datasets and reconstruction metrics.

1 INTRODUCTION

Creating high-quality 3D environments at scale remains a central challenge in computer vision and graphics, with direct practical consequences for film-making, VR/AR, and physical AI systems that rely on closed-loop simulation. In all of these settings, the useful output is not only a plausible image, but an explicit 3D representation that can be rendered in real time, queried from multiple viewpoints, and used for physically grounded interaction.

Recent neural 3D reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover such representations from posed images and have established strong reconstruction quality under controlled input conditions. Their empirical scalability, however, is bounded by the availability of accurate camera poses and high-quality multi-view images. The constraint is stronger for dynamic scenes, where synchronized multi-camera capture is typically required (Li et al., 2022). Reconstruction pipelines are also limited by the content already observed in the input views and therefore cannot reliably extrapolate to unobserved regions. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) reduce per-scene optimization costs, but they introduce another quantitative bottleneck: training requires diverse, large-scale 3D data, and data scarcity often appears as poor out-of-distribution generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a data-rich alternative. Because they are trained on massive internet-scale video corpora, they exhibit strong fidelity and broad generalization across scene types. Their training videos contain varied camera trajectories, including handheld motion, cinematic pans, and drone footage, so the models implicitly learn regularities about 3D structure without requiring curated multi-view supervision. Compared with reconstruction-only approaches (Charatan et al., 2024), generative models can also synthesize plausible content beyond the visible input frames. The limitation is representational: video diffusion models output 2D frames rather than explicit geometry, which prevents direct use in simulation and other downstream tasks that require geometric consistency, long-horizon coherence, and physical interaction. Recent camera-control results (Wang et al., 2024e; Ren et al., 2025) show that video models can be adapted with relatively small posed datasets to generate image sequences under specified camera trajectories. This makes them an attractive empirical source for 3D reconstruction: the generated posed sequences can serve simultaneously as inputs, supervision, and comparison targets.

Motivated by this observation, we connect reconstruction and generation through *generative 3D scene reconstruction*. We introduce *Lyra*, a method that generates explicit 3D environments from video-diffusion latents in a single forward pass. Crucially, *Lyra* is trained in a self-distillation framework (Fig. 2), where a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a video diffusion model. Given a single input image or video, we sample camera trajectories as conditioning inputs, denoise the corresponding video latents, and decode each latent through two aligned branches. The standard RGB decoder produces the teacher video sequence, while the 3DGS decoder produces the student explicit 3D representation. To make the supervision cover a broader range of viewpoints, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse their evidence while reducing long-term and multi-view inconsistencies.

This generative reconstruction design yields three measurable advantages: **(i)** it creates large-scale synthetic environments across diverse scenarios directly from video diffusion models, avoiding the collection of real-world multi-view captures; **(ii)** by operating in the video-model latent space, it aggregates many views more efficiently than pixel-space feed-forward reconstruction methods; and **(iii)** its explicit 3DGS output provides geometrically consistent renderable scenes suitable for downstream uses such as physical simulation. At inference time, *Lyra* therefore produces high-quality 3D Gaussian scenes from monocular input with real-time rendering support, *without requiring any additional optimization or post-processing*.

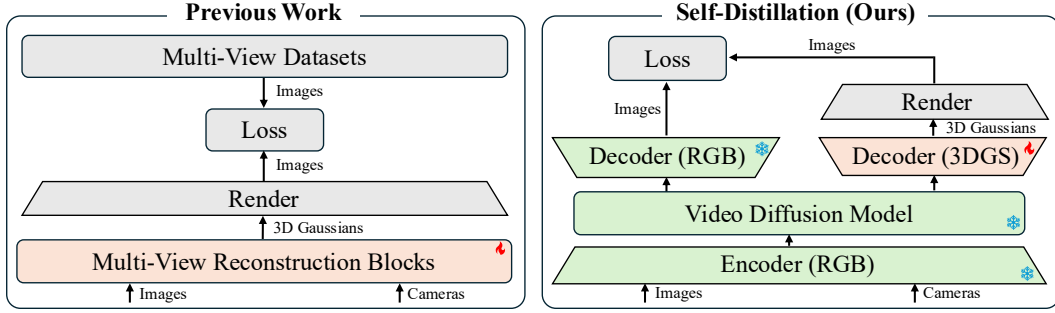


Figure 2: **Self-distillation framework of Lyra.** Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks with real-world datasets whose diversity is limited by captured multi-view data (Zhou et al., 2018; Ling et al., 2024b). Our self-distillation framework (right) instead uses the RGB decoder output of a pre-trained camera-controlled video diffusion model as the teacher and compares it with renderings from the 3DGS decoder student under matched camera trajectories. Because the teacher is pre-trained on diverse 2D video data, the student receives broad multi-view supervision without requiring captured multi-view scenes.

We further extend the same self-distillation framework to dynamic 4D generation from monocular video. In this setting, the video model supplies space-time supervision, and the student learns time-conditioned 3DGS representations that can render dynamic scenes from novel viewpoints.

Overall, our work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the need for captured multi-view real-world data.
- We extend the framework to dynamic scenes, enabling 4D reconstruction from a monocular video under the same teacher-student supervision principle.
- We show that the model generalizes across diverse scenes and achieves state-of-the-art quantitative results for both single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily evaluate object-centric scenes with limited or absent background content. Later work (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) expands the setting to scene-scale generation, making the comparison more relevant for environment synthesis. CAT3D (Gao et al., 2024b) adapts a pre-trained image diffusion model to generate multi-view images from either generated or real image inputs. A separate optimization stage then reconstructs those generated views into an explicit 3D representation, such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Related methods extend this multi-view generation paradigm to dynamic 3D scenes with space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Although these systems report impressive visual results, their explicit-3D output depends on expensive per-scene optimization that is not amortized across examples. Our comparison target is therefore different: instead of first generating multi-view images and then optimizing a reconstruction, we aim to *directly* generate a 3D scene from text or a single image.

Camera-conditioned video generation. Camera-controlled video diffusion has progressed rapidly as a way to evaluate generation under specified viewpoints. MotionCtrl (Wang et al., 2024e) introduced camera control by conditioning pre-trained video models on camera poses. Subsequent approaches (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) use Plücker coordinates for pixel-wise camera conditioning, while another group of methods (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) studies training-free camera control. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrate strong camera-controlled video synthesis by scaling synthetic training data. These methods provide useful camera-conditioned RGB outputs but do not themselves produce an explicit 3D representation. Our work is orthogonal to this line: we use a camera-controlled video model as a teacher for feed-forward 3D generation, so novel viewpoints can be rendered *after* generation. Concretely, we build on GEN3C (Ren et al., 2025) and use it as the teacher in a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction methods mainly focus on object-centric scenes without full scene backgrounds, directly predicting NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or image inputs. Recent scene-level models (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) regress per-pixel 3D Gaussians from one or more input images, improving speed but tying performance to the training distribution. In practice, many scene-level reconstruction models are trained on datasets such as RealEstate10K (Zhou et al., 2018), and their generalization to generated or out-of-distribution scenes remains limited. Recent methods adapt feed-forward reconstruction for scene generation. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps along multi-view images for feed-forward 3D reconstruction. Most closely related, Wonderland (Liang et al., 2025a) uses a camera-controlled video model and a feed-forward network to generate 3D Gaussians. In contrast, our method uses the camera-controlled video model as a self-distillation teacher, training the 3D student *without requiring real-world multi-view data*. We further show that the same setup extends to feed-forward 4D scene generation, a setting not addressed by these baselines.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our central design choice is to convert the implicit 3D evidence contained in video diffusion latents into an explicit 3DGS decoder that produces high-quality renderable 3D representations. We implement this conversion with a teacher–student framework: the video diffusion model acts as the teacher by generating RGB videos, and the 3DGS decoder acts as the student by predicting geometry in the same latent space, as illustrated in Fig. 2. This section first specifies the video diffusion backbone used for camera-controlled supervision (Sec. 3.1) and then describes how that backbone provides the supervision signal for self-distillation (Sec. 3.2). We then detail the 3DGS decoder design in Sec. 4 and describe the minimal changes needed to evaluate the same framework on dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised only by a camera-controlled video diffusion model, the teacher’s 3D consistency directly determines the quality of the student signal. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model designed to preserve scene structure under viewpoint changes. Below, we summarize the GEN3C components that are most important for our supervision pipeline.

Spatiotemporal 3D cache. GEN3C improves both video consistency and camera-control accuracy by constructing a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos. Each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimate (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The resulting cache is organized as an $L \times V$ array, where L is the temporal length and V is the number of camera views.

Rendering and structured guidance. GEN3C uses this cache by rendering each point cloud from arbitrary target camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . For a camera sequence $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements yields V videos of length L , which are stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify pixels that the video diffusion model should synthesize rather than copy from the cache. These rendered RGB/mask pairs therefore provide structured visual guidance with an explicit camera condition for the later teacher outputs.

Video variational autoencoder (VAE). Diffusion-based video generation models commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress high-resolution videos into a lower-dimensional latent space, reducing training and inference cost. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, the pre-trained VAE encoder $\mathcal{E}$ maps the video to $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. The diffusion model is trained and evaluated in this compressed latent space rather than in pixel space. The final RGB video is recovered as $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ using the pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we use the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Its latent channel dimension is $C = 16$, its temporal dimension is $L' = (L - 1)/\tau + 1$, and its

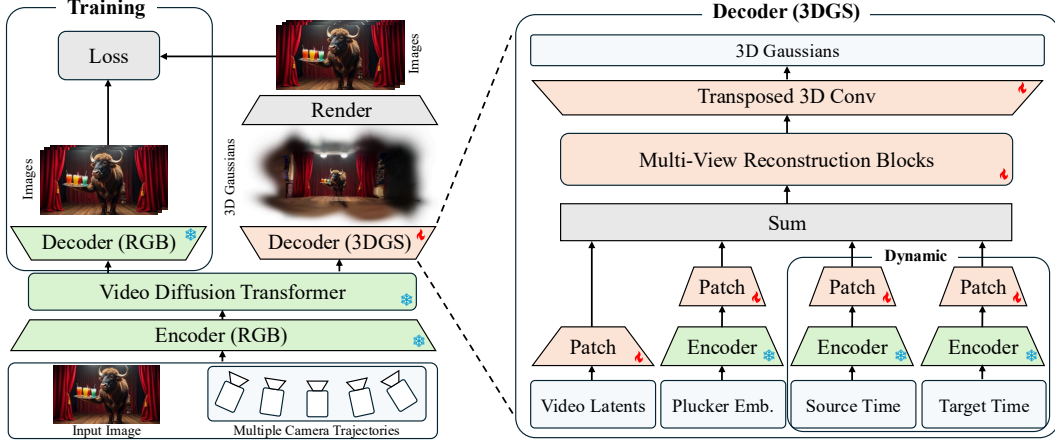


Figure 4: **3D Generative Reconstruction Framework.** The pipeline uses a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data as a fixed teacher. We train only the 3D Gaussian Splatting (3DGS) decoder by matching rendered 3DGS views to the RGB-decoded teacher generations under the same camera trajectories, while the autoencoder and diffusion model remain frozen. At inference time, the 3DGS decoder is used directly and the RGB decoder is no longer required. Adding time conditioning to the 3DGS decoder extends the same comparison and supervision structure from static to dynamic 3D scene generation.

spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as the student in a teacher–student pipeline where the camera-controlled video diffusion model $\mathcal{V}$ supplies the supervision. To make the supervision distribution broad, we curate a large-scale prompt set with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and then expand each image into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion teacher $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded through two matched branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ generates teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, and the 3DGS decoder $\mathcal{D}_s$ predicts explicit 3D Gaussians $\mathbf{G}$. Student renderings are computed as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised against the teacher RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$. This aligned comparison gives the student a direct metric target under the same camera condition as the teacher output.

Multi-trajectory supervision. To increase viewpoint coverage for each input image, we sample $V = 6$ camera trajectories, as shown in Fig. 3. For every trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ produces latents $\mathbf{z}^v$, and the teacher decodes them into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse the six trajectory latents $\mathbf{z}^v$ into a coherent Gaussian scene $\mathbf{G}$ while completing disoccluded regions. This multi-trajectory self-distillation trains $\mathcal{D}_s$ entirely from synthetic supervision provided by $\mathcal{V}$; the latent-space design enables efficient aggregation across many views, and the explicit Gaussian output $\mathbf{G}$ supplies the geometry needed for consistent rendering and simulation.

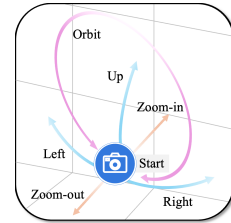


Figure 3: Six sampled camera trajectories provide broader viewpoint coverage for each input.

4 FEED-FORWARD

RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ maps the synthesized multi-view latents from the video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We use 3D

Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it provides an explicit scene representation together with fast rendering. Fig. 4 (right) visualizes the decoder architecture.

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Prior feed-forward reconstruction frameworks typically generate 3D Gaussians per pixel, so their practical input scale is limited by the number and resolution of views. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, and AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video teacher produces $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, a scale that exceeds the GPU-memory capacity of existing pixel-space approaches. The limiting factor is attention over visual tokens, whose memory and compute grow with the number of pixels. We therefore avoid pixel-space scaling and instead operate directly on the compressed video latent space produced by the camera-controlled video diffusion model. The resulting multi-view video latents are denoted $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps the multi-view latent inputs into the hidden dimension of the reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both represented at latent resolution $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input is first patchified to match the reconstruction-block hidden dimension. No additional visual encoder is required; a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) converts the video latents into flattened reconstruction tokens. The summed latent and camera tokens are processed with the reconstruction blocks. Following Long-LRM (Ziwen et al., 2024), each block contains one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers, and we repeat the block twice to obtain 16 layers with 512 hidden dimensions. A final transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. We compute raw Plücker embeddings from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. To align camera information with the video latent representation, we reuse the pre-trained VAE RGB encoder $\mathcal{E}$. Specifically, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately using $\mathcal{E}$, then concatenate the resulting latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 so that it matches the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. The primary student–teacher comparison is an image-based reconstruction objective between rendered 3DGS views and RGB-decoded teacher frames. We decompose this objective into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lips}$ (Zhang et al., 2018) computed with VGG (Simonyan & Zisserman, 2014) features.

Depth supervision. RGB-only supervision can produce flattened geometry, even when image-space reconstruction metrics are reasonable. To improve the geometric evidence available during training, we additionally supervise rendered depth maps with consistent video depth estimated by ViPE (Huang et al., 2025), using the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we regularize opacity with an L1 penalty $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity, yielding a more compact rendered representation.

Total loss. The total training objective is the weighted sum of the image, perceptual, depth, and opacity terms with weights λ_i :

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lips} \mathcal{L}_{lips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

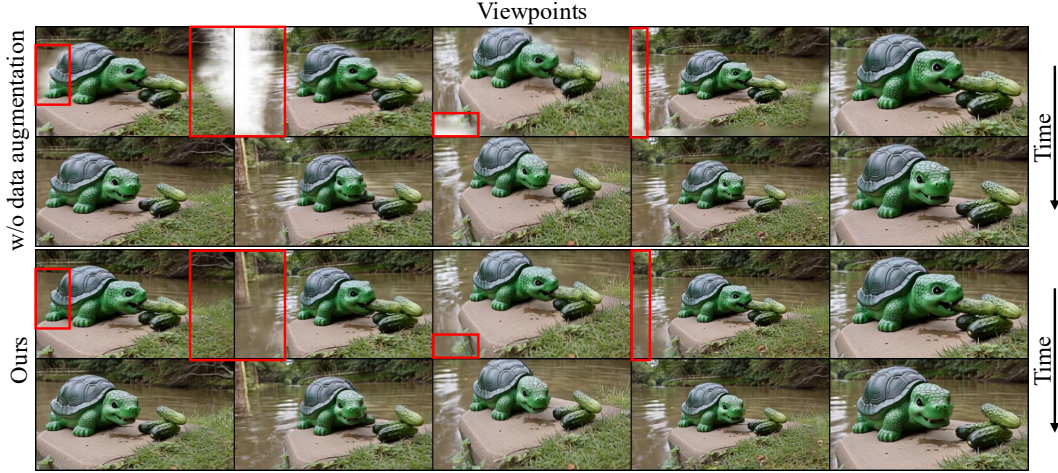


Figure 5: **Dynamic data augmentation.** The naive time-aware 3DGS decoder receives uneven supervision across timestamps, and early timesteps under extreme poses show low-opacity artifacts in regions outside the available supervision. Our motion-reversed augmentation gives each timestep broader spatial coverage, reducing these low-opacity artifacts under the same dynamic rendering condition.

5 EXTENSION TO DYNAMIC 3D SCENES

The same self-distillation pipeline extends to dynamic 3D scenes with only small architectural and data changes.

Self-distillation for dynamic scenes. The dynamic setup follows the static 3D design while replacing the single input image with a monocular input video of length L and corresponding camera poses. The video model then generates multi-view video latents that preserve the same underlying motion across viewpoints. For video inputs, we use the same data-generation pattern as in the static setting: diverse text prompts are sampled with large language models, videos are generated using video diffusion models (Cosmos, 2025; Wan, 2025), and the generated videos are annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and define the dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all video frames. Architecturally, the dynamic 3DGS decoder $\mathcal{D}_s$ matches $\mathcal{D}_s$ except that the $\mathcal{D}_s$ input is augmented with encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times are assigned to each input frame, giving $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, while the raw target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated, augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, and encoded with the RGB encoder $\mathcal{E}$ to obtain $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, we randomly sample a target time step, decode the corresponding 3D Gaussians, and supervise them across all trajectories produced by dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$, initializing the patchification layers for $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zeros. The encoded time inputs $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. For static 3D scenes, time does not affect the underlying geometry, so frames from different timesteps can all supervise the same 3DGS renderings. Dynamic scenes require a stricter comparison: because the 3DGS changes over time, only frames at the corresponding timestamp provide valid supervision. This reduces the number and diversity of supervision views at each time and can lead to a trivial solution in which the 3DGS at one timestep ignores evidence from other timesteps (see Fig. 5).

To address this imbalance, we introduce dynamic data augmentation that gives each timestep both near and far supervision views. Early frames naturally correspond to viewpoints near the input, whereas later frames are farther away. We counter this bias by pairing supervision views for every timestep. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$,

Table 1: **State-of-the-art comparisons.** Single-image-to-3D results on RealEstate10K, DL3DV, and Tanks-and-Temples show that our method obtains the best reported PSNR, SSIM, and LPIPS in every dataset–metric comparison.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

which produces six additional multi-view sequences. After reversing those sequences back to the original motion order, we obtain six inward trajectories that move from far viewpoints toward the input. Together with the original six outward trajectories, this provides 12 supervision views per timestep. The augmentation is used only during training: the reversed trajectories serve as additional supervision to prevent collapse from pruning artifacts and are not needed at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. Our training setup is designed to test whether self-distilled video supervision can replace captured multi-view datasets. We therefore do not train on any existing multi-view dataset; instead, we construct the *Lyra* dataset using only our video model to supervise the 3DGS decoder. The static 3D reconstruction split contains 59,031 images, and the dynamic 4D split contains 7,378 videos. All examples are generated from diverse text prompts covering indoor and outdoor scenes, humans, animals, realistic content, and imaginative content. For each image or video, we synthesize six camera trajectories, yielding 354,186 videos for 3D training and 44,268 videos for 4D training.

Baselines. We compare against state-of-the-art generative single-image-to-3D methods: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, the main benchmark comparison uses the quantitative results reported in the corresponding papers.

Evaluation. We follow prior work and evaluate single-image to 3D generation on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). The evaluation protocol matches Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b), allowing direct comparison under the same dataset and task conditions. We report the standard reconstruction metrics PSNR, SSIM, and LPIPS, where higher PSNR/SSIM and lower LPIPS indicate better novel-view reconstruction quality.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports the single-image-to-3D comparison across RealEstate10K, DL3DV, and Tanks-and-Temples using PSNR, SSIM, and LPIPS. Our method achieves the best result in every reported dataset–metric cell, outperforming all listed prior methods under the same benchmark protocols. The pattern is consistent across both in-domain-style real-estate data and broader scene benchmarks, indicating that the self-distilled 3DGS decoder benefits from the teacher’s generative coverage while retaining reconstruction accuracy. Since the student is supervised by the camera-controlled video model, further improvements in video-generation quality and consistency should translate directly into stronger 3D reconstructions. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Fig. 1 and Fig. 6 show novel-view renderings from the generated 3DGS scenes; video results will be released upon acceptance. The qualitative examples complement the metrics by showing that the model renders unseen regions with high visual quality while remaining consistent with the input image or video. Additional qualitative comparisons on the *Lyra* dataset are included in Appendix C.



Figure 6: **Image-to-3DGS Generation.** Five rendered viewpoints from each generated 3DGS scene illustrate the same multi-view consistency measured by the quantitative novel-view metrics.



Figure 7: **Ablations.** Each variant is compared by rendering the same extreme novel viewpoint after image-to-3DGS generation, making artifacts and consistency changes directly comparable across design choices; depth visualizations are provided in Appendix C.4.

6.3 ABLATIONS

Tab. 2 isolates the contribution of each design choice on out-of-distribution diverse prompts. All variants are evaluated by comparing 3DGS renderings against videos generated by the camera-controlled video diffusion teacher, so each ablation uses the same supervision target, metrics, and evaluation condition.

Real data only. When the student is trained only on real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), performance drops substantially relative to our self-distilled model. This comparison highlights the generalization gap of commonly used real datasets on out-of-distribution generated scenes.

Self-distillation + real data. Adding real data to the self-distillation training set does not improve over self-distillation alone. The near-matched PSNR and lower SSIM/LPIPS performance indicate that the synthetic teacher supervision is already diverse and consistent enough to train the reconstruction model without additional captured multi-view data.

No depth loss. Removing depth loss reduces image-based metrics and produces flatter geometry. Depth supervision therefore improves geometric plausibility while also preserving, and slightly improving, the reported PSNR, SSIM, and LPIPS results. We visualize depth maps in Appendix C.4.

No opacity pruning. Keeping all Gaussians instead of pruning by opacity makes the representation less compact and slightly lowers visual

Table 2: **Ablation study on the *Lyra* dataset.** Each row changes one data, loss, or architecture component and is evaluated with PSNR/SSIM (higher is better) and LPIPS (lower is better) against the same video-teacher rendering targets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

quality. The practical runtime difference is also clear: rendering at $H = 704$ and $W = 1280$ takes 18ms with pruning versus 30ms without pruning, corresponding to a $1.67\times$ speed up.

No LPIPS loss. The LPIPS term improves robustness to teacher-output inconsistencies and better preserves high-frequency detail. Quantitatively, removing it causes one of the largest LPIPS degradations in the ablation results, confirming that perceptual supervision is important for the visual quality targeted by the benchmark.

No multi-view fusion. A naive alternative is to generate 3D Gaussians independently for each camera trajectory and then merge the resulting point clouds. The ablation shows that this strategy is much weaker than learned multi-view fusion: allowing reconstruction tokens from different trajectories to attend to one another produces substantially better PSNR, SSIM, and LPIPS.

No Mamba-2. Replacing the joint Transformer/Mamba-2 design with Transformer-only blocks yields slightly lower reconstruction quality and a much slower forward pass. For $V = 6$, $L = 121$, $H = 704$, and $W = 1280$, one forward pass takes 3213ms with the joint blocks compared with 20922ms using Transformer-only blocks, a $6.5\times$ speedup for the proposed design.

No latent-based 3DGS. Pixel-space reconstruction does not scale to the supervision volume produced by the teacher. Previous pixel-space methods such as BTimer (Liang et al., 2025b) are restricted to 12 input frames, whereas our latent-space decoder can use up to 726 input frames. Running the 3DGS decoder in pixel space instead of latent space therefore leads to out-of-memory, making latent-based reconstruction a practical requirement rather than only an architectural preference.

7 CONCLUSION

We presented *Lyra*, a 3D and 4D generation framework that reconstructs explicit 3DGS scenes from a single image or video. Rather than collecting captured multi-view datasets, the method uses a camera-controlled video diffusion model as a teacher and trains a student 3DGS decoder through self-distillation. The decoder operates in the video latent space and directly predicts 3D Gaussians, avoiding per-scene optimization and post-processing. Across the reported experiments, this design yields state-of-the-art quantitative results for static and dynamic scene generation, with consistent improvements across datasets, metrics, and ablated design choices. The current scale and consistency of generated scenes are still bounded by the capacity of the camera-controlled video diffusion teacher. Improving that teacher should therefore provide a direct path to larger and more consistent 3D scene synthesis. Another promising direction is adapting auto-regressive techniques (Chen et al., 2024a) to extend our framework to even larger environments. Finally, modeling motion and tracking information inside the reconstruction network, as explored in concurrent work (Lin et al., 2025), could further improve dynamic-scene motion quality.

8 ETHICS STATEMENT

Like other generative AI technologies, our method carries misuse risks, including the creation of misleading or synthetic 3D/4D content. At the same time, the quantitative and practical motivation of this work is to improve scalable simulation, robotics, and embodied AI by providing controllable tools for data generation. We expect the main positive impact to come from the following capabilities and empirical insights.

- **The ability to enable realistic simulation for embodied agents:** The method produces generated 3D and 4D environments with controllable camera motion and physically consistent viewpoints, supporting evaluation and training of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By removing the requirement for multi-view capture and expensive per-scene optimization, the method supports high-fidelity, customizable scene generation at a scale relevant to robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** As an academic study, the work provides a concrete case study of how camera control, spatiotemporal generative models, and explicit 3D

reconstruction interact, which can inform future research in computer vision, graphics, and embodied AI.

We caution against deceptive or wrongful use of realistic generative outputs, particularly when media could be misleading or difficult to verify. Safeguards such as provenance tracking, dataset documentation, and careful evaluation practice can help ensure that these models are used in ways that advance science and society.

9 REPRODUCIBILITY STATEMENT

To support full reproduction of the reported results and accelerate follow-up comparisons, we will release the training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section gives implementation details for the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1, with emphasis on the modifications that improve the reliability of its 3D supervision signal.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage: object parts that are occluded in the source view, such as the lion body in Fig. 8, can become visible from a novel viewpoint without being marked correctly in the disocclusion mask. In the visual comparison, this failure appears as incomplete foreground completion by the video diffusion model in Fig. 8 (second and fourth columns).

To make the teacher signal more conservative under occlusion, we assume that regions with unknown geometry in $\mathbf{P}^{t,v}$ may be occupied and derive refined disocclusion masks from that assumption. This encourages the video model to synthesize potentially visible foreground regions that standard point-based rendering would miss.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This mesh defines a surface boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained by ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ is the original forward-warping disocclusion mask and ϵ is a small tolerance factor. If the mesh surface is closer than the original points, we mask the region as a potential foreground disocclusion. This produces more reliable guidance for video generation under large viewpoint changes, as shown by the third and fifth columns of Fig. 8.

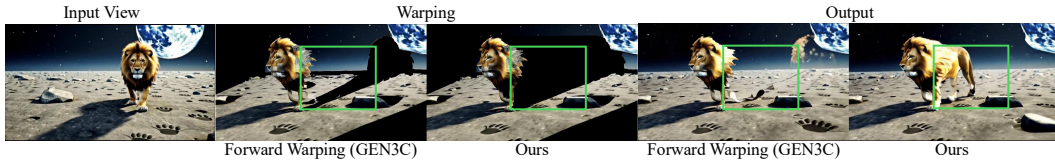


Figure 8: **Rendering function comparison.** Our model builds on GEN3C (Ren et al., 2025), which uses forward-warped images as camera-control conditioning. Comparing forward warping (left) with our conservative rendering function (right) shows that the highlighted region is incorrectly filled with background pixels by forward warping but is masked by our method, enabling the video model to complete the foreground correctly.

B MODEL DETAILS

Progressive training. Training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is computationally expensive, so we use the progressive schedule reported in Tab. 3. The final static pre-trained model initializes the dynamic model, allowing the dynamic stage to start from an already learned reconstruction prior. End-to-end training takes 6 days on 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Pixel-aligned 3D Gaussian generation scales linearly with spatial and temporal resolution. Our generated scenes cover 726 frames at 704×1280 resolution, corresponding to 654,213,120 per-pixel Gaussians if represented densely. We therefore generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the count by a factor of 64 to 10,222,080. Opacity-based

Table 3: **Progressive training setup.** The schedule increases resolution, sequence length, and view count before the dynamic stage, allowing the final model to train on high-resolution multi-trajectory supervision.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

pruning further reduces the representation to 2,044,416 Gaussians, making the output substantially more compact while preserving the reported rendering quality.

Joint Transformer-Mamba blocks. We adopt the reconstruction block design of Long-LRM (Ziwen et al., 2024), where one block consists of one Transformer layer followed by seven Mamba-2 layers. Repeating the block twice gives 16 layers with 512 hidden dimensions. This is smaller than the networks used in Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. Although pure Transformers can be expressive, they are substantially slower than Mamba2 during training and inference. Because both configurations were trained under the same budget, we hypothesize that Transformer-only blocks would require many more iterations to close the remaining rendering-quality gap.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we encode cameras with Plücker embeddings. These embeddings are computed at full spatial and temporal resolution and then projected into the video latent space with the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels and Plücker embeddings are 6-dimensional, we encode the 3-dimensional ray directions separately from the 3-dimensional cross product of ray directions and origins. The two latent encodings are then concatenated along the channel dimension.

Time encoding. To make the 1-dimensional time input compatible with the 3-channel RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to each raw time value. The resulting 3-channel time representation is replicated spatially and encoded into the compressed video latent space. This lets the decoder compare source and target time information in the same latent coordinate system used for video and camera features.

Dynamic data augmentation. Fig. 9 visualizes the dynamic augmentation for two example camera trajectories. The procedure creates paired supervision views that observe the same motion state from different extreme viewpoints, improving spatial coverage for each timestamp.

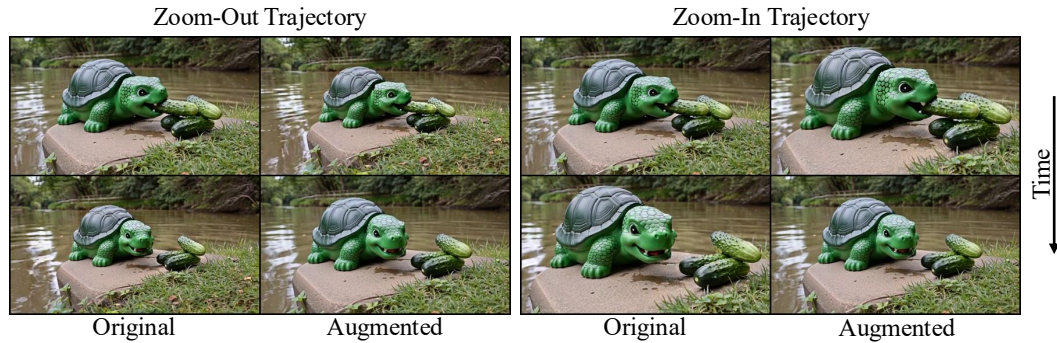


Figure 9: **Dynamic data augmentation videos.** Motion-reversed videos expand the supervision coverage for each timestamp, so early and late motion states are compared from both near and far viewpoints. We show zoom-out and zoom-in trajectories together with their augmented videos, whose camera motion is flipped.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) by rendering five views from the same generated scenes. The visual comparison matches the quantitative static evaluation and shows fewer artifacts with higher fidelity for our method.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare with BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. To ensure a fair baseline on the same scenes, we asked the BTimer authors to run experiments using the *Lyra* evaluation scenes. Because BTimer is regression-based, we integrate it with the same GEN3C (Ren et al., 2025) video diffusion backbone used by our method. Specifically, we use the same camera trajectories and provide RGB-decoded GEN3C videos as BTimer inputs. BTimer operates in pixel space and requires 12 input images. As discussed in the main paper, pixel-space models such as BTimer run out of memory when using all 726 high-resolution GEN3C frames, so we uniformly subsample the generated frames. For static evaluation, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluation, we sample 11 frames across the viewpoint range and include the bullet-time frame corresponding to the target motion state, ensuring that the reconstructed motion state is always present as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 and Tab. 4a compare our method with BTimer (GEN3C). The comparison uses the same generated scenes and standard PSNR, SSIM, and LPIPS metrics, and our method shows higher fidelity with fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare with BTimer (GEN3C), described in Sec. C.1, on 100 out-of-distribution videos from the dynamic *Lyra* dataset. We crop videos to 512×512 for a fair comparison because BTimer was primarily trained at that resolution. Tab. 4b reports the PSNR, SSIM, and LPIPS results and shows a clear advantage for our method over BTimer (GEN3C).

Table 4: **Quantitative results on static and dynamic *Lyra* datasets.** Compared with the baseline, our method improves PSNR and SSIM while reducing LPIPS in both the static and dynamic evaluation settings.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Static *Lyra* dataset comparison. (b) Dynamic *Lyra* dataset comparison.

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depth maps from generated 3DGS scenes. The comparison shows that depth supervision reduces flat geometry while maintaining visual quality in the rendered views.

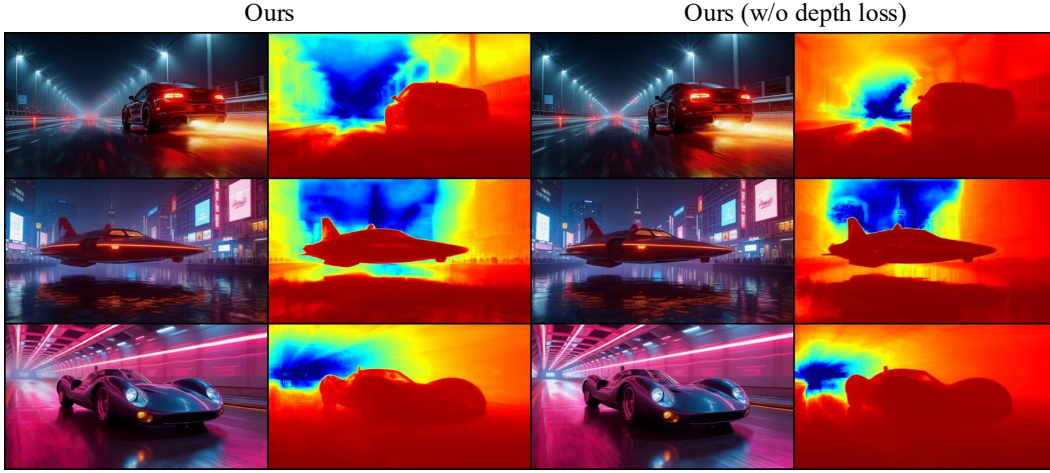


Figure 11: **Depth loss ablation.** We compare 3DGS renderings with their corresponding depth maps. Adding depth supervision prevents flat geometry while preserving the rendered-view quality measured in the main ablation table.

D ROBOT SIMULATION

A primary practical motivation for diverse 3D generation is scalable simulation for autonomous agents. We therefore build an end-to-end export pipeline: first, our model generates 3D scenes from text; second, the 3D Gaussians are exported as a .ply file; third, the .ply files are converted to .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create the .usdz files. The resulting .usdz scene can be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. These generated environments can then support training and testing AI-based robots under diverse conditions. Early demo results will be released upon acceptance, and Fig. 12 shows a screenshot of the interface.

E RELATED WORK

This section provides additional related-work context and clarifies the comparison with CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b), summarized visually in Fig. 13.

3D generation. Early 3D generation methods mainly focused on single object categories, extending GANs to 3D by using neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With CLIP-based supervision (Radford et al., 2021), the field moved toward more flexible asset generation and editing from text (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). Diffusion-based approaches later improved visual quality by replacing CLIP guidance with Score

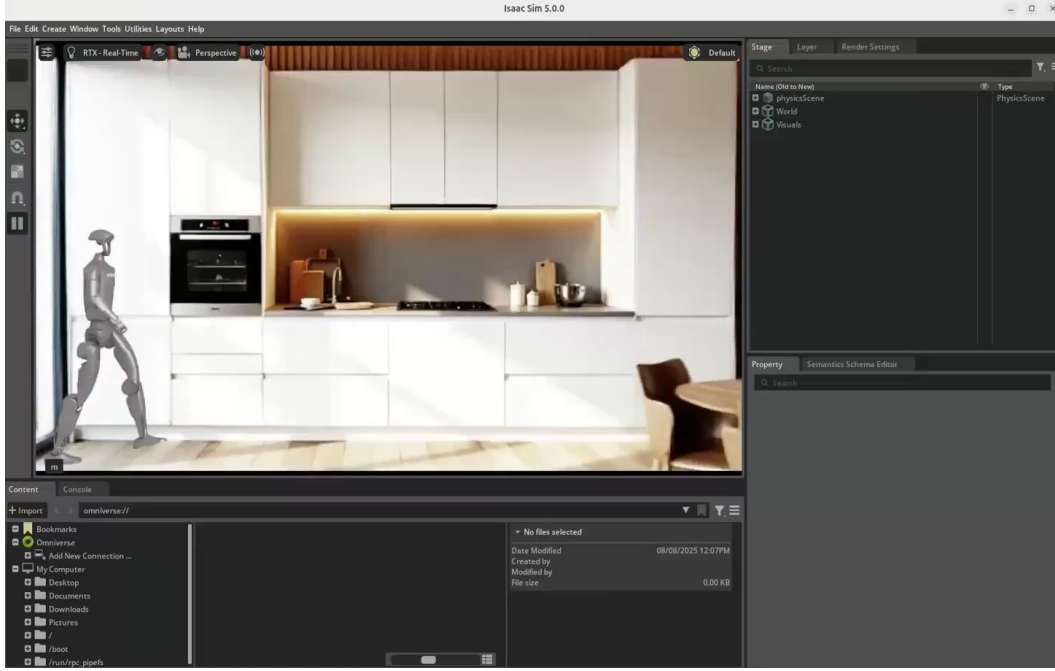


Figure 12: **Robot Simulation.** A generated 3DGS scene from our method is imported into Isaac Sim 5.0, illustrating the practical use of the reconstructed representation as a simulation environment for embodied-agent evaluation.

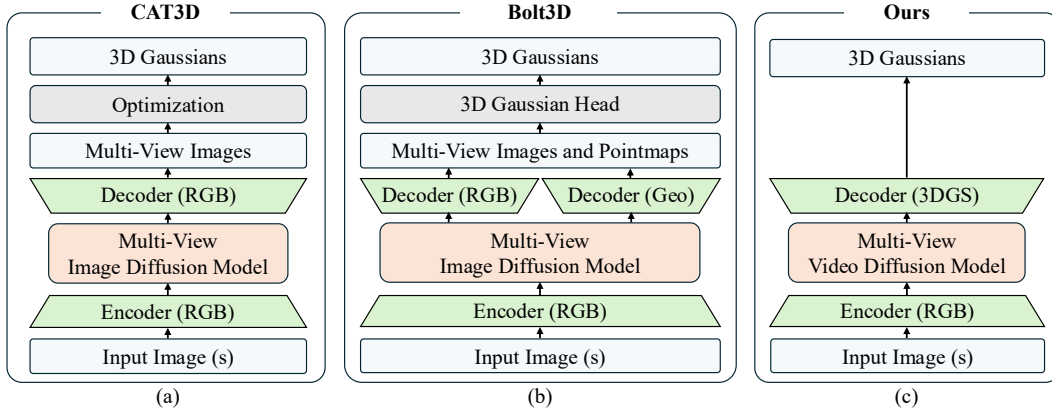


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) first generates novel-view images with a multi-view image diffusion model and then optimizes those images into 3D Gaussians. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps through a geometry autoencoder and predicts the scene with a feed-forward 3D Gaussian head rather than per-scene optimization. Our method uses a pre-trained camera-controlled video diffusion model and *directly* decodes multi-view video latents into 3D Gaussians, making the comparison point the latent-to-3D reconstruction step.

Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To improve structural coherence, many methods enforce multi-view consistency by generating or constraining scenes from multiple perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b); others synthesize scenes through iterative inpainting (Höllein et al., 2023; Shriram et al.,

2024). A related line lifts 2D images into 3D with NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or meshes combined with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Feed-forward 3D methods aim to reduce reconstruction time by directly predicting 3D content from images or text (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Compared with our method, most of these approaches are restricted to static 3D scenes. Other methods target specific domains such as faces (Kirschstein et al., 2025). Concurrent approaches (Liang et al., 2025b; Xu et al., 2025) address real scenes but do not handle diverse generated scenes or large viewpoint changes as targeted here.

4D generation. 4D generation, or dynamic 3D scene synthesis, has advanced quickly, with many methods using text prompts or images as conditioning signals. After early large-scale generative models for this task (Singer et al., 2023), later work improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, many methods lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other methods add physics priors (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), improve motion controllability with templates (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), or emphasize compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another research direction extends 3D GAN frameworks to 4D by training on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). These methods are often limited by small-scale, single-category datasets, which restricts generalization. They also tend to be object-centric and may neglect background elements, whereas our method leverages recent video generation models to target broader scene-level photorealism.

Camera-conditioned video models. Camera control in video diffusion models has become an important way to evaluate generation under explicit viewpoint conditions. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Later work (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) improved conditioning with Plücker-coordinate cameras, while other methods follow MotionCtrl by using flattened poses (Wang et al., 2025b). Training-free control methods (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) avoid additional trainable parameters. Most of these approaches remain based on U-Net architectures, while low-rank camera-control methods are also explored (Zhang et al., 2025b).

Recent U-Net-based camera-control work continues to improve viewpoint controllability (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), and several methods explore diffusion-transformer backbones. Cheong et al. (Cheong et al., 2024) propose a transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control but uses pre-defined, non-continuous camera trajectories. Other methods (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) study pose and geometry estimation with video DiTs. CAT4D (Wu et al., 2025b) fine-tunes a multi-view backbone into a multi-view video diffusion model. Large-scale synthetic data further improves camera-controlled video generation in SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), and 4DiM (Watson et al., 2024) trains a space-time diffusion

1566 model from scratch for single-image novel-view synthesis. Another direction retargets camera motion
1567 in videos with underlying dynamics (Seo et al., 2025; Ma et al., 2025). Finally, spatiotemporal
1568 caches based on point clouds or optical flow have become a common conditioning mechanism for
1569 camera-controlled video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al.,
1570 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al.,
1571 2025).

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